Benhao Huang

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA, USA **2025.7 - 2026.12 (Expected)**
MS in Machine Learning (Advisor: Prof. Zico Kolter)

Shanghai Jiao Tong University, Shanghai, China **2021 - 2025**
B.Eng. in Computer Science and Technology (IEEE Honor Class)

- **GPA: 93.14/100.00, 4.08/4.30**

RESEARCH INTERESTS

I aim to make foundation models more capable and efficient through **Pretraining** and **Model Architecture Design**, focusing on architectures that use **Compute and Memory** better and on the **Optimization Dynamics** that govern how models learn and scale. Most recently, I study looped (weight-tied) language models and their fixed points, aiming for deeper reasoning with lower memory and faster inference.

SELECTED PUBLICATIONS

[1] Towards Looped Models Done Right: Part II: Rethinking at Fixed Points

Preprint [Paper] [Code] [Tweet]

***Benhao Huang**, Chufan Shi, Junlin Chen, Shicheng Wen, Zhengzhong Liu, Eric Xing, Xuezhe Ma*

- Showed that once looped LMs converge near fixed points, the recurrence path can be skipped, so compute scales without memory.
- Exploited this with three shortcuts: terminal KV sharing cuts the decoding KV cache **3**×, distilled prefill runs up to **1.79**× faster on 8K prompts, and reusing saved rollout states halves RL scoring and backward time.
- Proposed a learned depth prior (trained from prediction feedback with an entropy term) and orthogonal input injection; from **100M to 1.6B** parameters, both beat Huginn's prior and Parcae's injection in perplexity, and at 1.6B the learned prior matches full-KV fixed-depth training with a 3× smaller KV cache.

[2] Towards Looped Models Done Right: Part I: Topology, Input Injection, Recurrent-State Design

Preprint [Blog] [Code] [Tweet]

***Benhao Huang**, Chufan Shi, Junlin Chen, Shicheng Wen, Zhengzhong Liu, Eric Xing, Xuezhe Ma*

- Conducted apples-to-apples, from-scratch ablations of recurrence topology, input injection, and recurrent-state design at the 730M-parameter, **336B-token scale**, identifying the main contributors to performance gains; the resulting model outperformed the recurrent baseline across all 10 benchmarks.
- Transferred these design findings to MoE models at **8B** total/0.8B active parameters and **500B** training tokens; under matched training and inference FLOPs, the resulting looped MoE used **75% fewer resident parameters** than a 112-layer feedforward MoE while outperforming it on DROP and GSM8K and matching it on MATH500.

[3] Equilibrium Reasoners: Learning Attractors Enables Scalable Reasoning

ICML 2026 [Paper] [Code] [Tweet]

***Benhao Huang**, Zhengyang Geng, Zico Kolter*

- Proposed an attractor-based framework for iterative reasoning, **casting reasoning as convergence to task-conditioned fixed points** in a latent dynamical system.
- Demonstrated reliable two-axis test-time scaling via depth (solver steps) and width (stochastic restarts), boosting accuracy from **8% to over 99%** on Sudoku-Extreme, enabled by **training interventions that reshape attractor landscapes** and inject noise for stable convergence.

[4] Flow Equivariant World Models: Structured Dynamics Outside the Field of View

ICML 2026 [Paper] [Code] [Website]

Hansen Lillemark, **Benhao Huang***, Fangneng Zhan, Yilun Du, T. Anderson Keller† (*Equal Contribution)*

- Introduced **FloWM**, an action-conditioned video world model that **predicts partially observable dynamics** within its memory representation using **flow and self-motion equivariance**.
- Outperformed DFoT and SSM on long-horizon video prediction, maintaining consistent memory of dynamics over 200 frames.

[5] PAN: A World Model for General, Interactable, and Long-Horizon World Simulation

Technical Report [Paper] [Website]

Jiannan Xiang, Yi Gu*, Zihan Liu*, Zeyu Feng*, Qiyue Gao*, Yiyan Hu*, **Benhao Huang***, Guangyi Liu*, Yichi Yang*, Kun Zhou*, + 20 others, Hector Liu†, Zhiting Hu†, Eric P. Xing† (*Core Contributor)*

- **Co-led large-scale data curation, filtering, and captioning pipelines**; accelerated early **training and evaluation**; influenced key architecture and experiment choices.
- PAN is a **large-scale interactive** video world model for **open-domain, action-conditioned simulation**, combining an autoregressive backbone for **long-horizon temporal consistency** with a diffusion-based decoder for **short-term fidelity**, enabled by sliding-window causal chunk-wise attention.
- Achieved **state-of-the-art performance** on planning, prediction, and controllability benchmarks, outperforming prior world models, including V-JEPA, Cosmos, and WAN, by up to **20%**.

[6] DCA-Bench: A Benchmark for Dataset Curation Agents

KDD 2025 (Oral) [Paper] [Code]

***Benhao Huang**, Yingzhuo Yu, Jin Huang, Xingjian Zhang, Jiaqi Ma†*

- Introduced a task for LLM agents: **detecting dataset quality issues to automate AI training data curation.** We collected 221 instances and designed four difficulty levels for fine-grained analysis of frontier models.
- Developed an LLM-based automatic evaluator for scalable evaluation, shown to be **robust to self-preference and length bias**.

HONOURS AND AWARDS

- **Outstanding Graduate, Shanghai Jiao Tong University** (**2025**)
- **National Scholarship,** ***Highest honor for undergraduates*** (**2024**)

LEADERSHIP AND SERVICES

Reviewer, ICML 2026, NeurIPS 2025, KDD 2025

Student Mentor, Shanghai Jiao Tong University, CS2612 and CS2601, 2023

Volunteer, Shanghai Marathon, 2022 - 2024

SKILLS

Programming: Python, PyTorch, C/C++, TorchXLA, Slurm

Languages: English (**TOEFL**: 107, S24, R27, L29, W27, **GRE**: V157, Q170, AW4.0)